ENTOMOLOGY

FOR 2 PERCUSSIONISTS/SPEAKERS,
SAMPLER,
WIND QUINTET,
AND PIANO

Notes by Composer
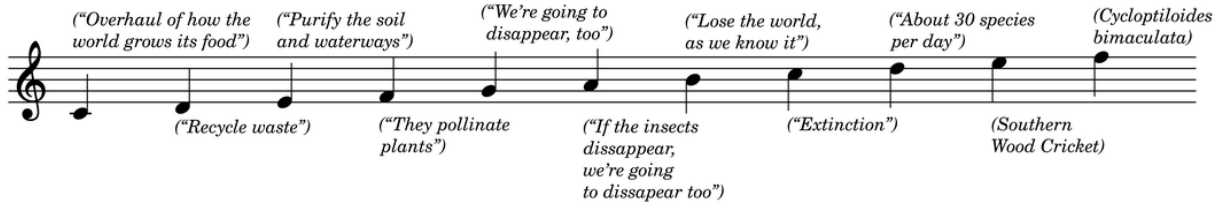
Entomology is focused on the decline of insect populations as part of the ongoing Holocene extinction. The first section is somewhat nostalgic, focusing on the childlike wonder of insects. Using graphic notation and serialism, the intense population decline is conveyed, switching to a more distressed tone color. The piece returns to the A section, this time with a more empty and isolated melody. The final section focuses more on the tools that are available in reversing this decline, from alternative pesticides to recent legislative support for various insect protections.

The samples used in are as follows:

1. Audio of *Cycloptiloides bimaculata*
2. Audio of *Southern Wood Cricket*
3. "About 30 species per day…Extinction": An excerpt of an interview of conservation biologist Axel Hochkirch, on insect extinction.
4. "Lose the world as we know it": An excerpt of an interview of entomologist Sónia Ferreira, on insect extinction.
5. "We're going to disappear, too": An excerpt of an Aljazeera interview of Paul Ralph Ehrlich.
6. A collection of excerpts of journalist Mereana Hond speaking on insect extinction and decline.
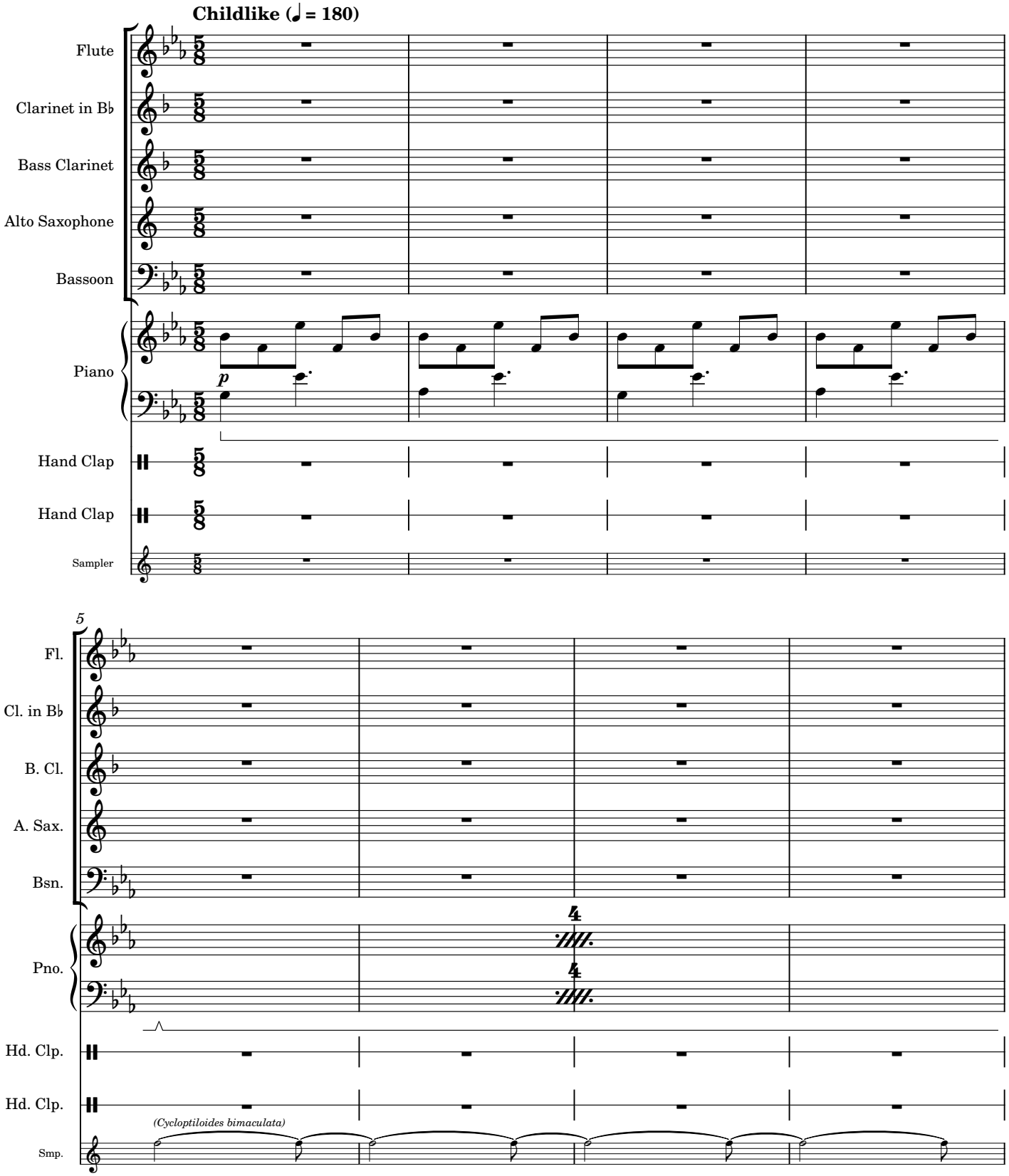
Performance Notes
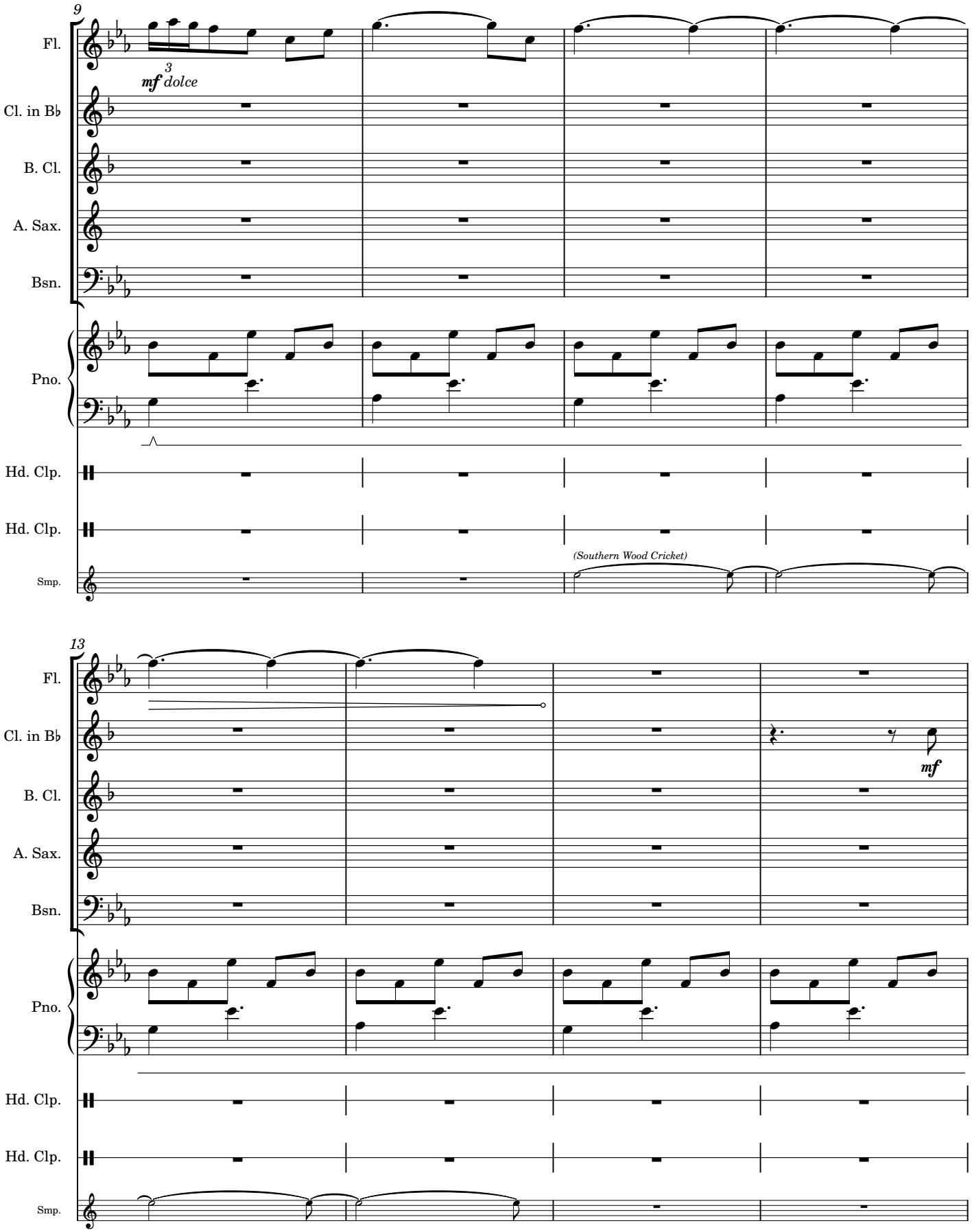
Entomology uses a sampler keyboard to enable the performance of excerpts from interviews and documentaries. The mapping used in the score is provided in the key below. Alternatively, if a sampler is unavailable, the part may be played through software such as Abelton Live or through a tape.
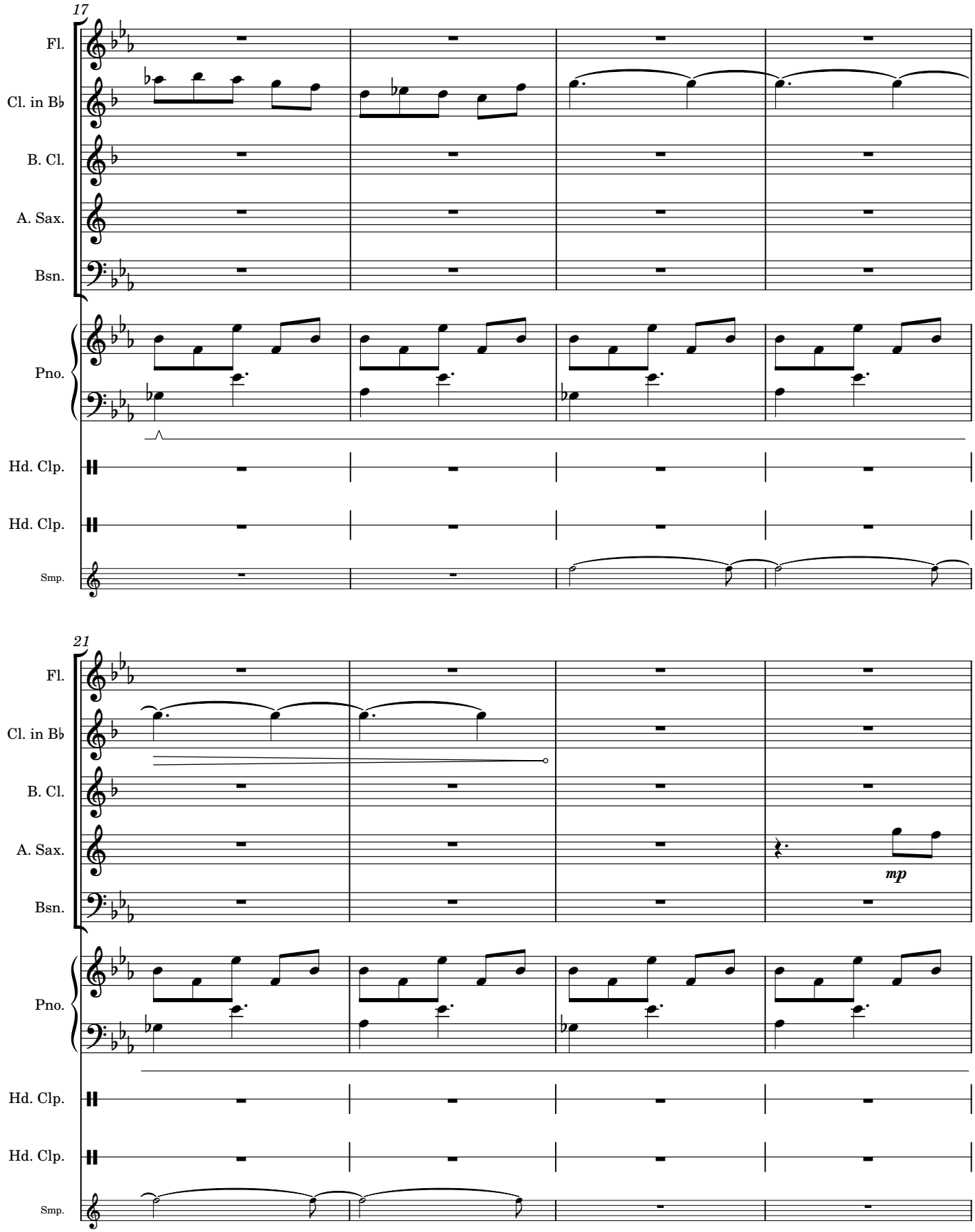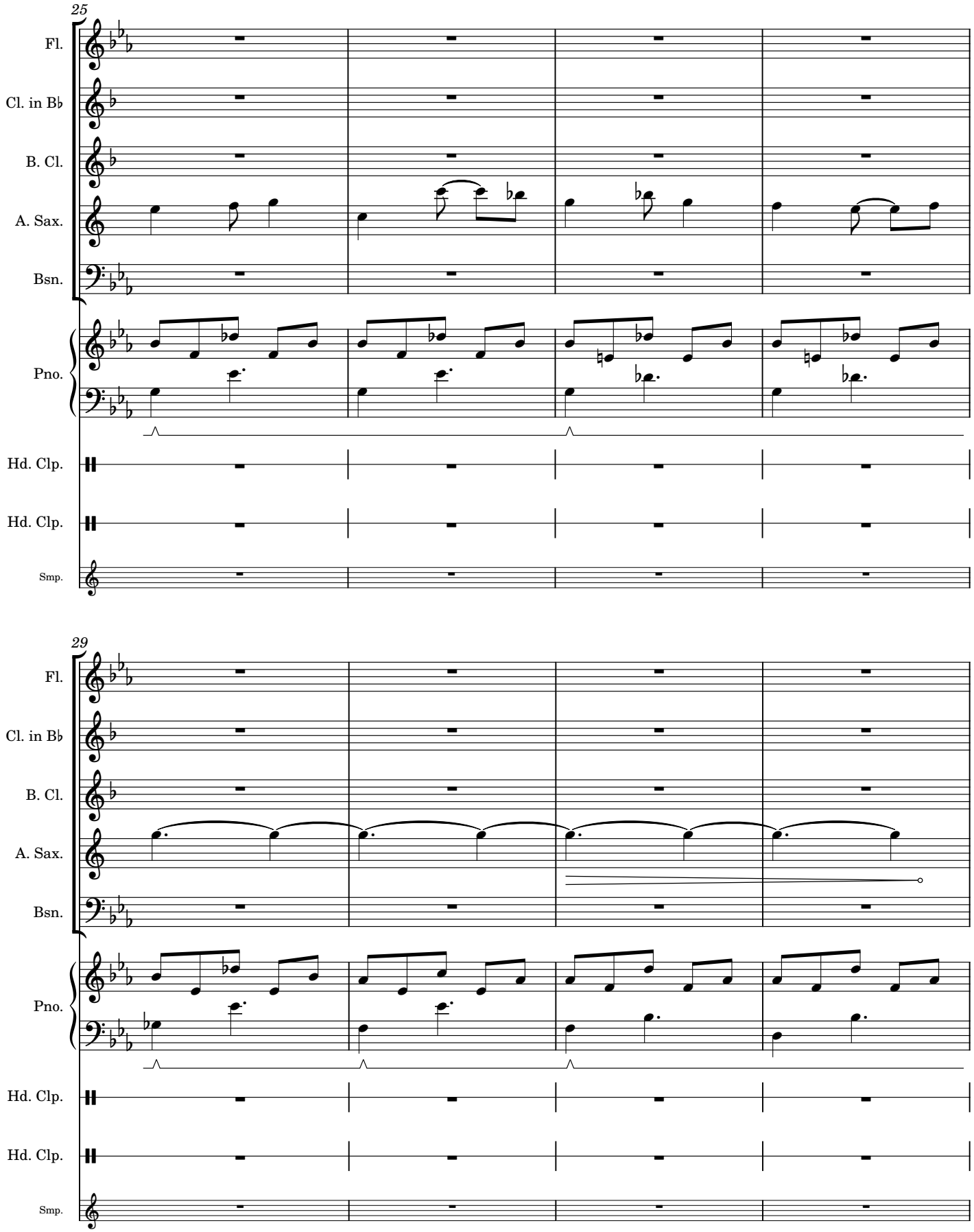


During the graphic sections in the score, performers should tend to more melodic phrases during ascending sections and dissonant phrases during descending sections.
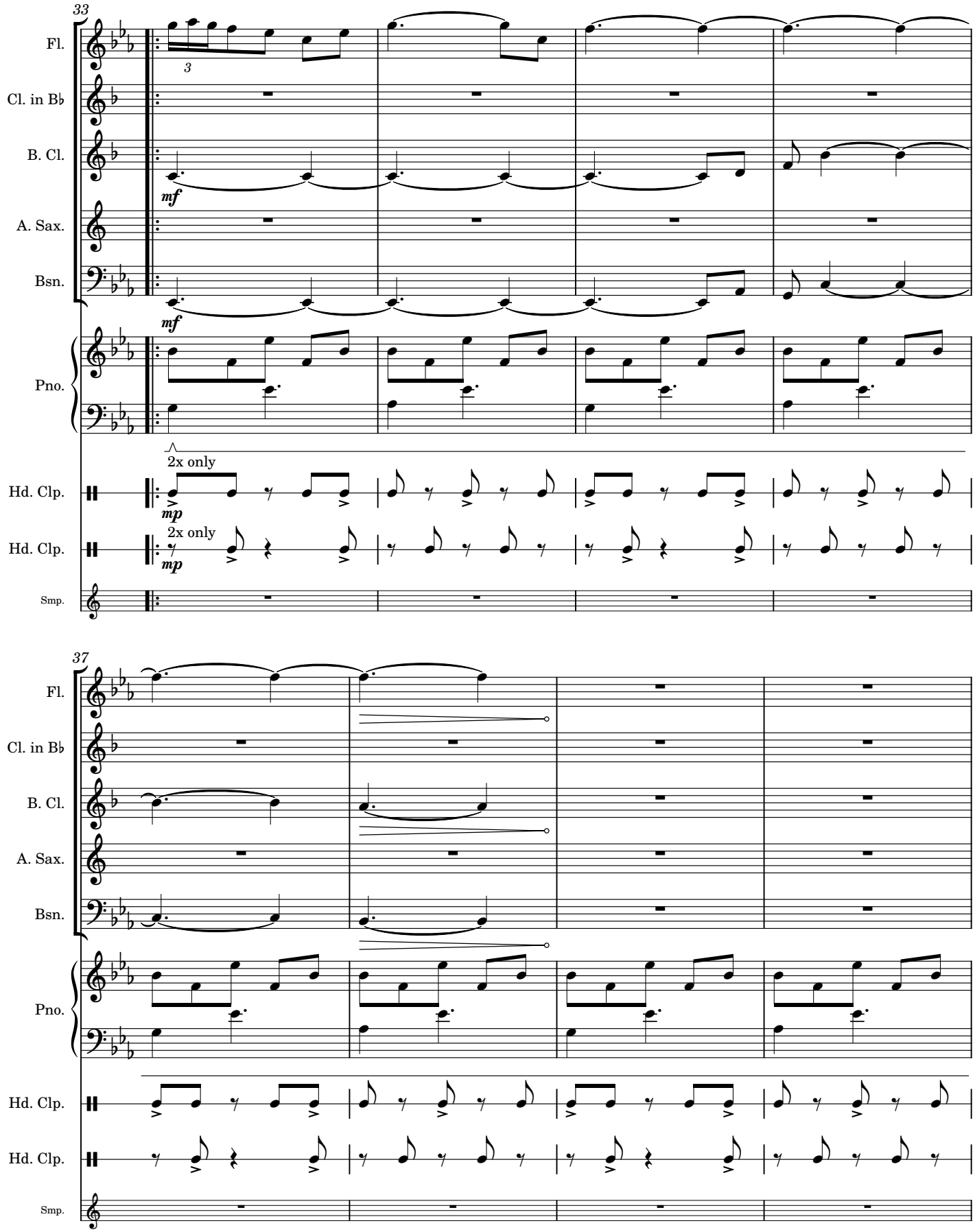
The 2 percussionists also double as speakers, calling out insect names before their population decline is played musically. During those sections, they should speak into microphones and amplified, in order to be heard over the ensemble. The first percussionist should also clap with a deeper tone using cupper hands, similar to the flamenco clap *palmas sordas*. The second percussionist should tend to clap with a brighter tone by striking one set of fingers against the opposite palm, similar to the flamenco clap *palmas claras*.
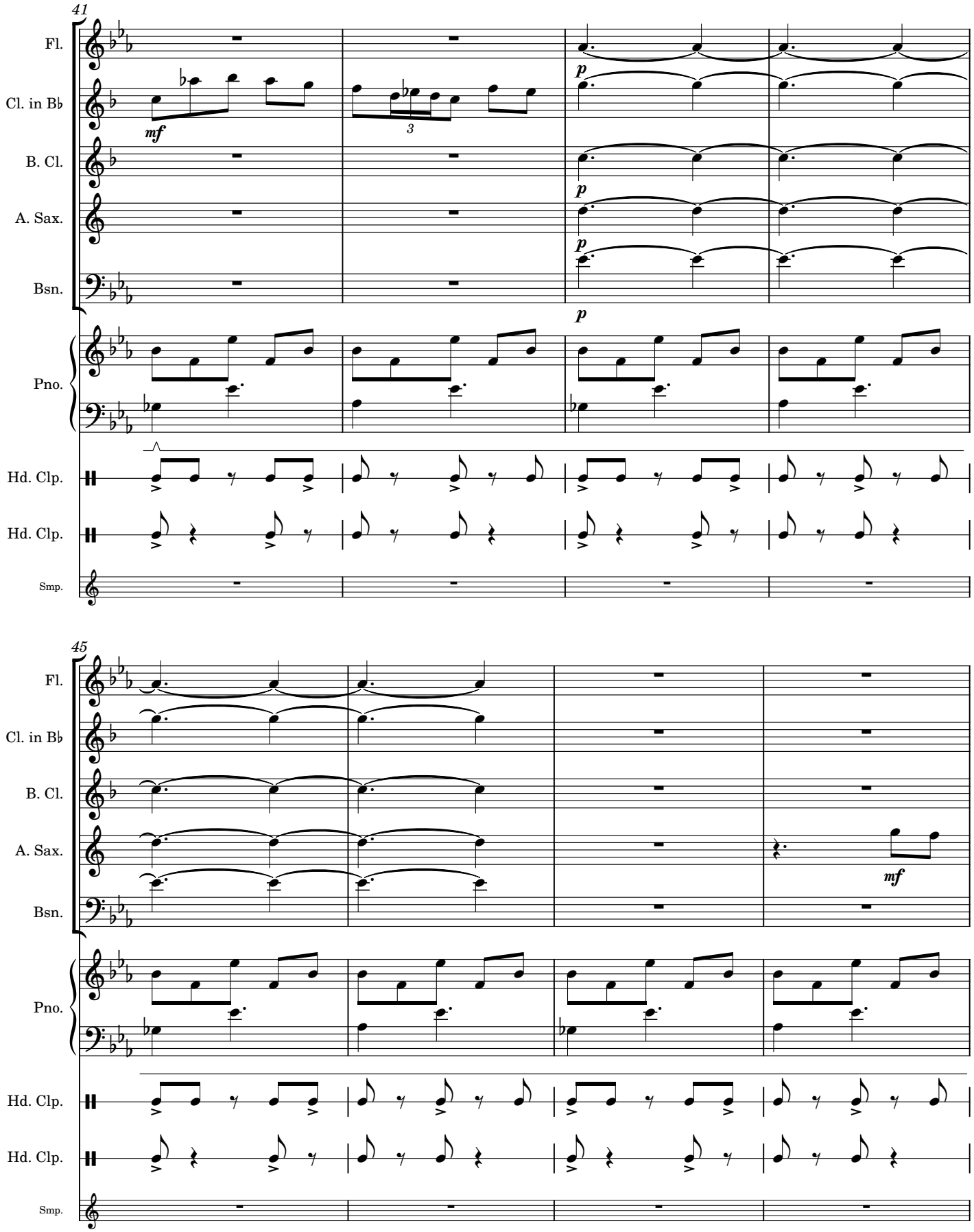

Childlike (♩ = 180)
Flute
Clarinet in B♭
Bass Clarinet
Alto Saxophone
Bassoon
Piano
p
Hand Clap
Hand Clap
Sampler
5
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
(Cycloptiloides bimaculata)
Smp.
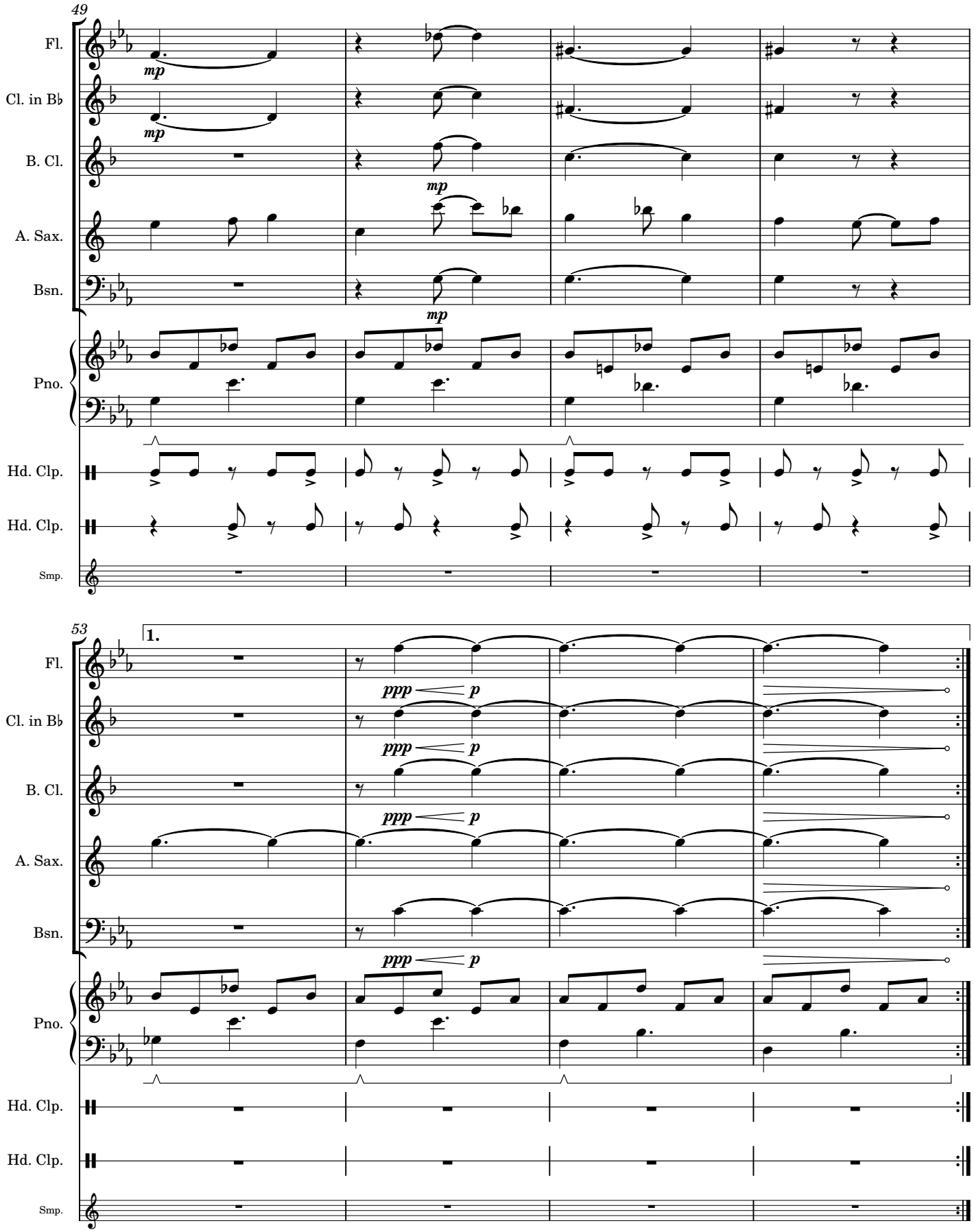

9
Fl.
3
mf dolce
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
(Southern Wood Cricket)
Smp.
13
Fl.
Cl. in B♭
mf
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
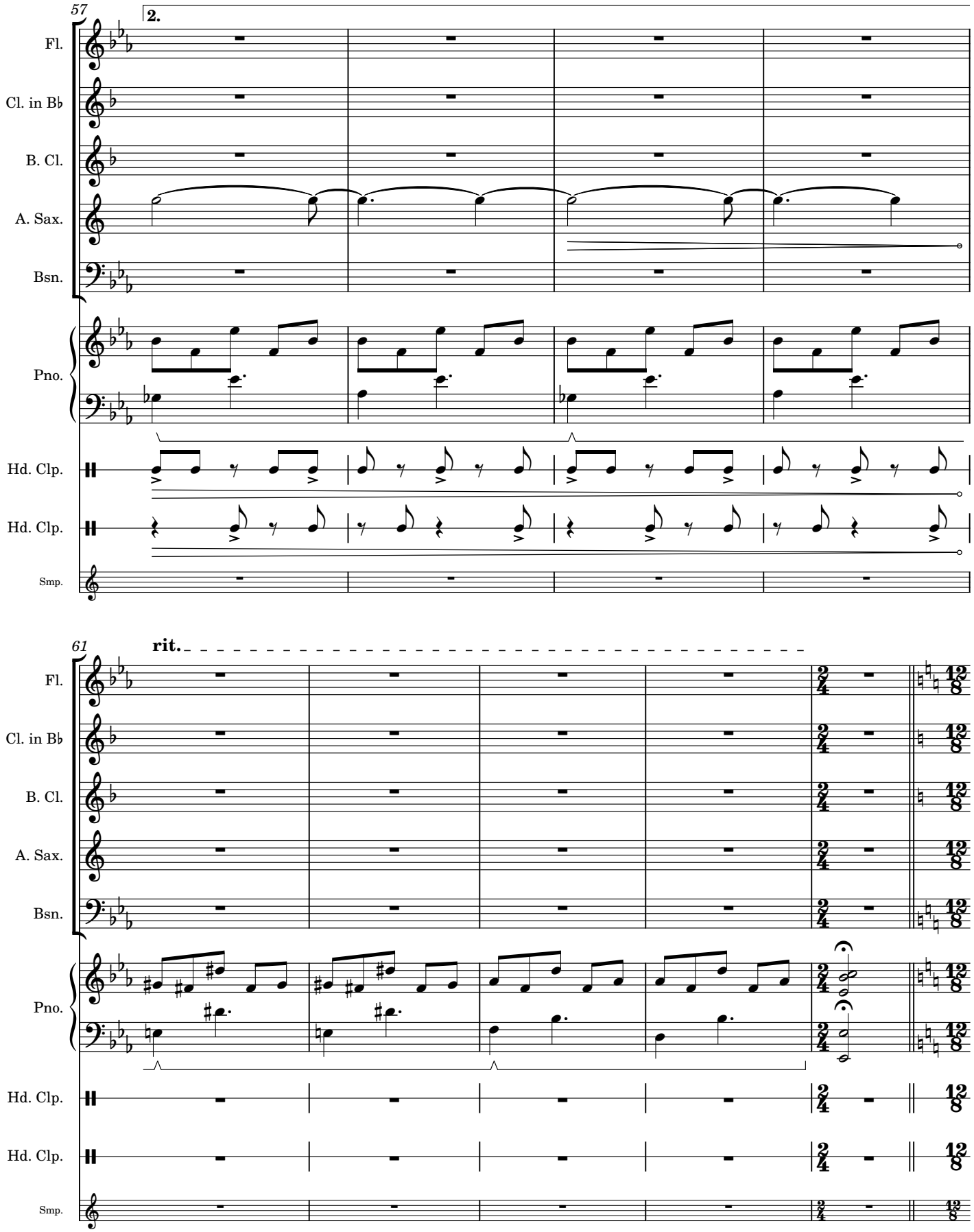

17
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
21
Fl.
Cl. in B♭
B. Cl.
A. Sax.
mp
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
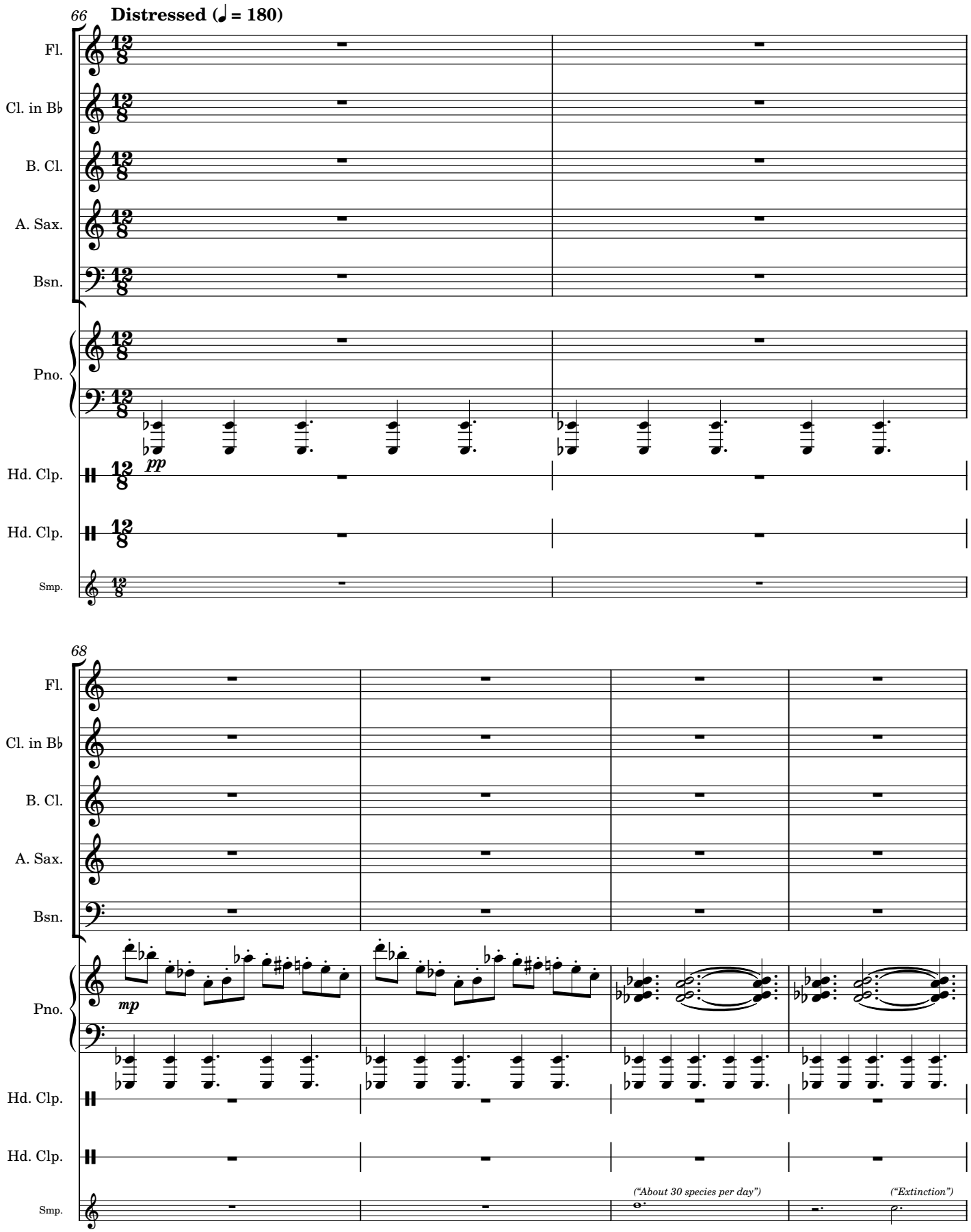

25
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
29
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
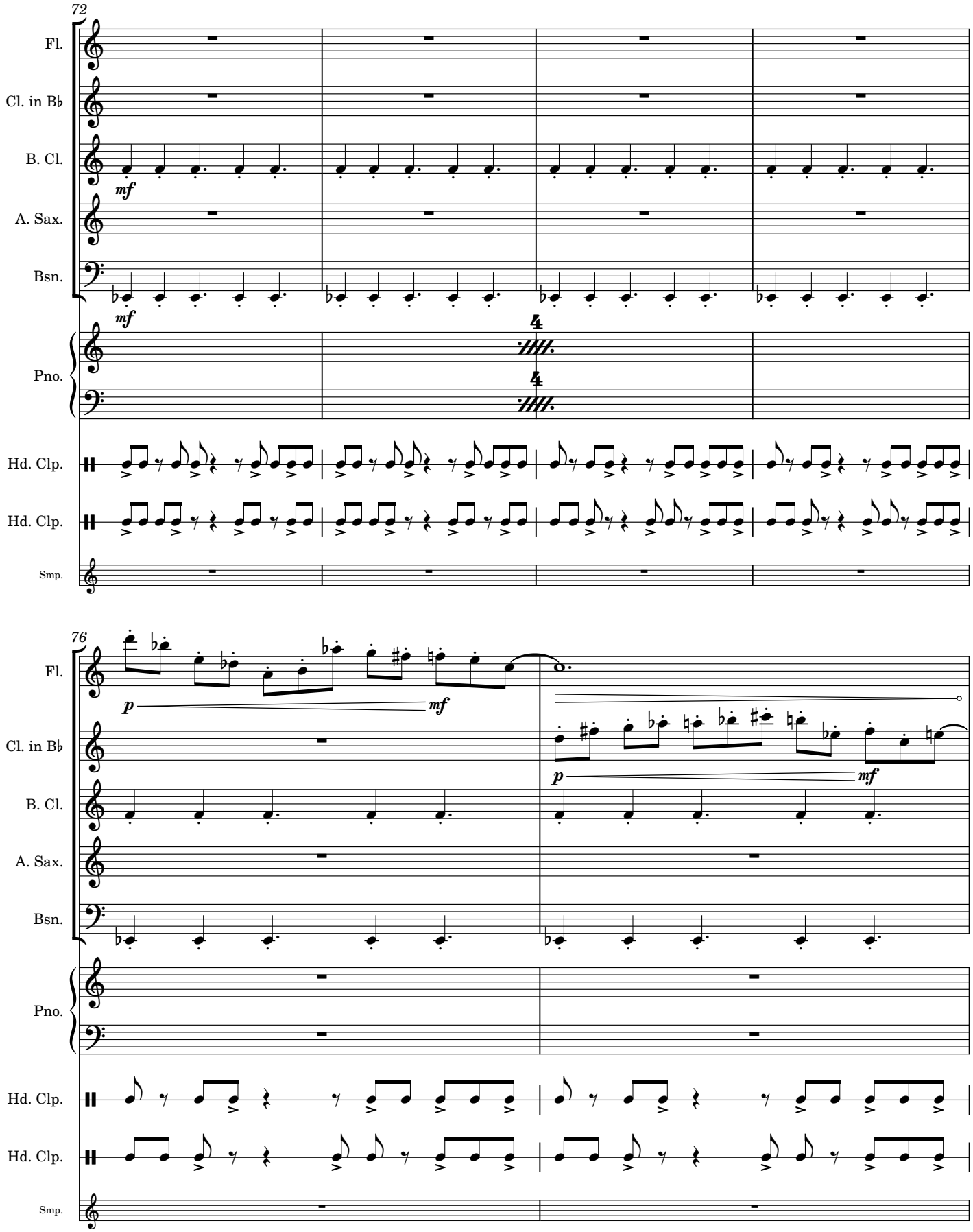

33
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
mf
Pno.
mf
2x only
Hd. Clp.
mp
2x only
Hd. Clp.
mp
Smp.
37
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
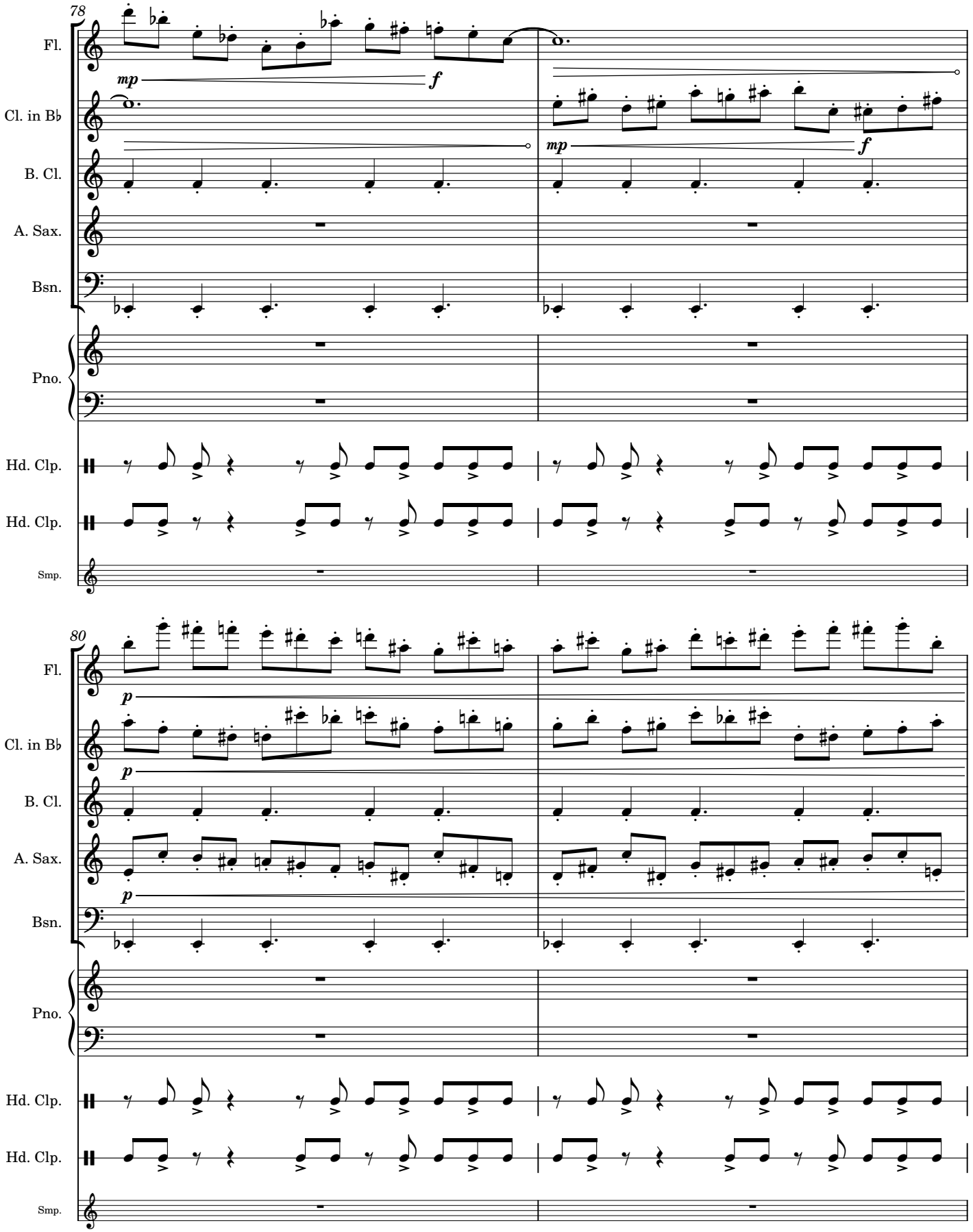

41
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mf
p
3
45
mf
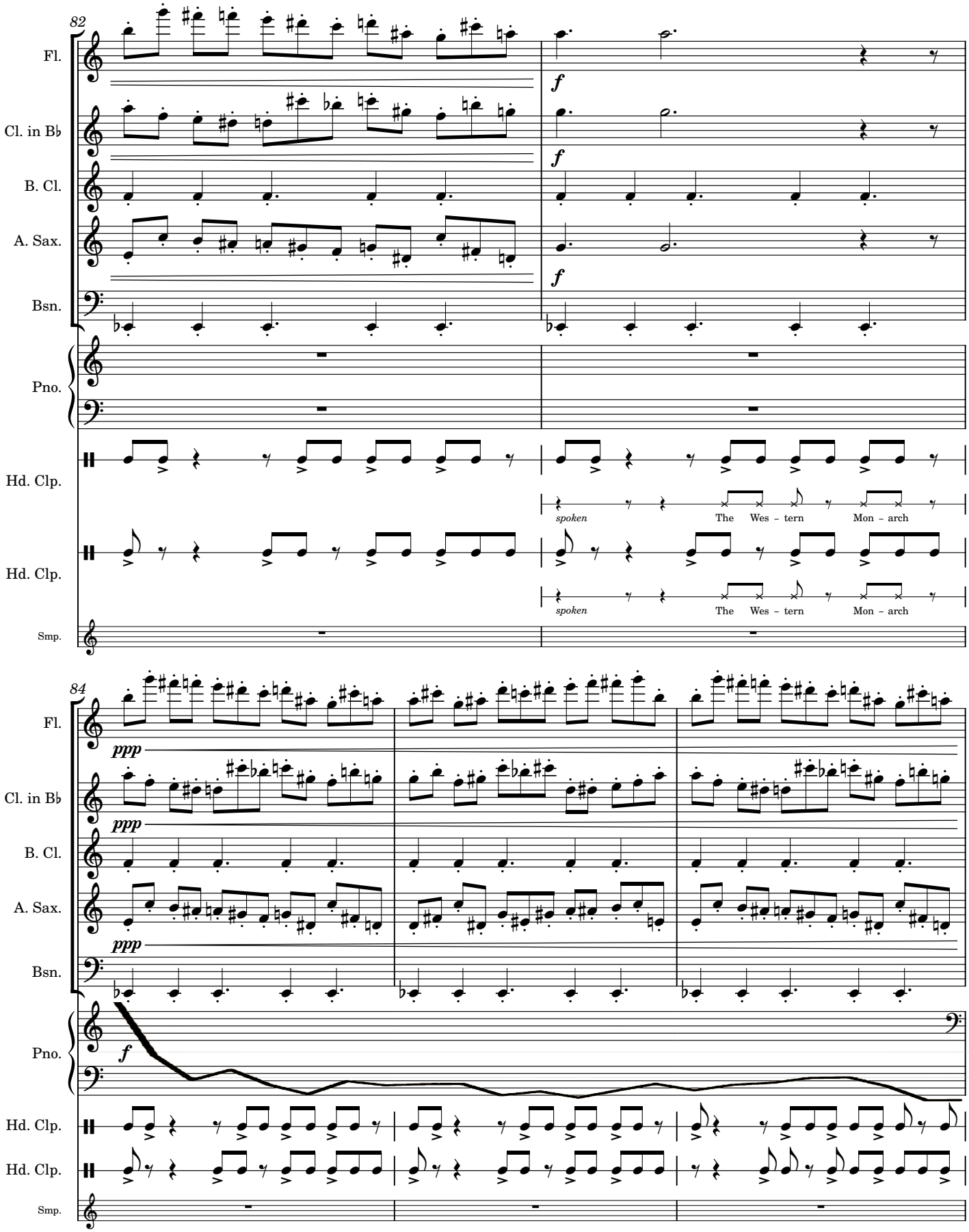

49
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mp
53
1.
ppp
p
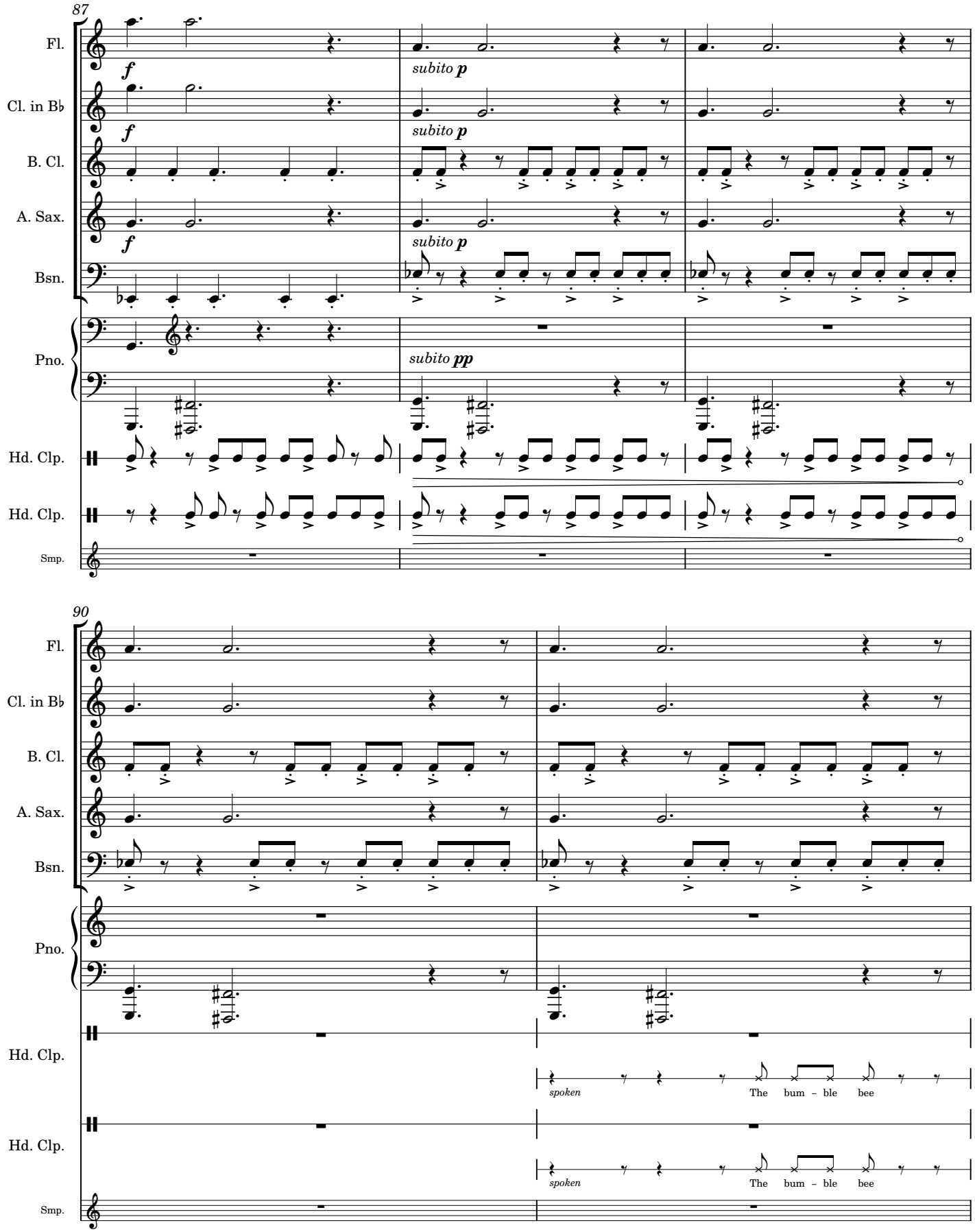

57
2.
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
61
rit.
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
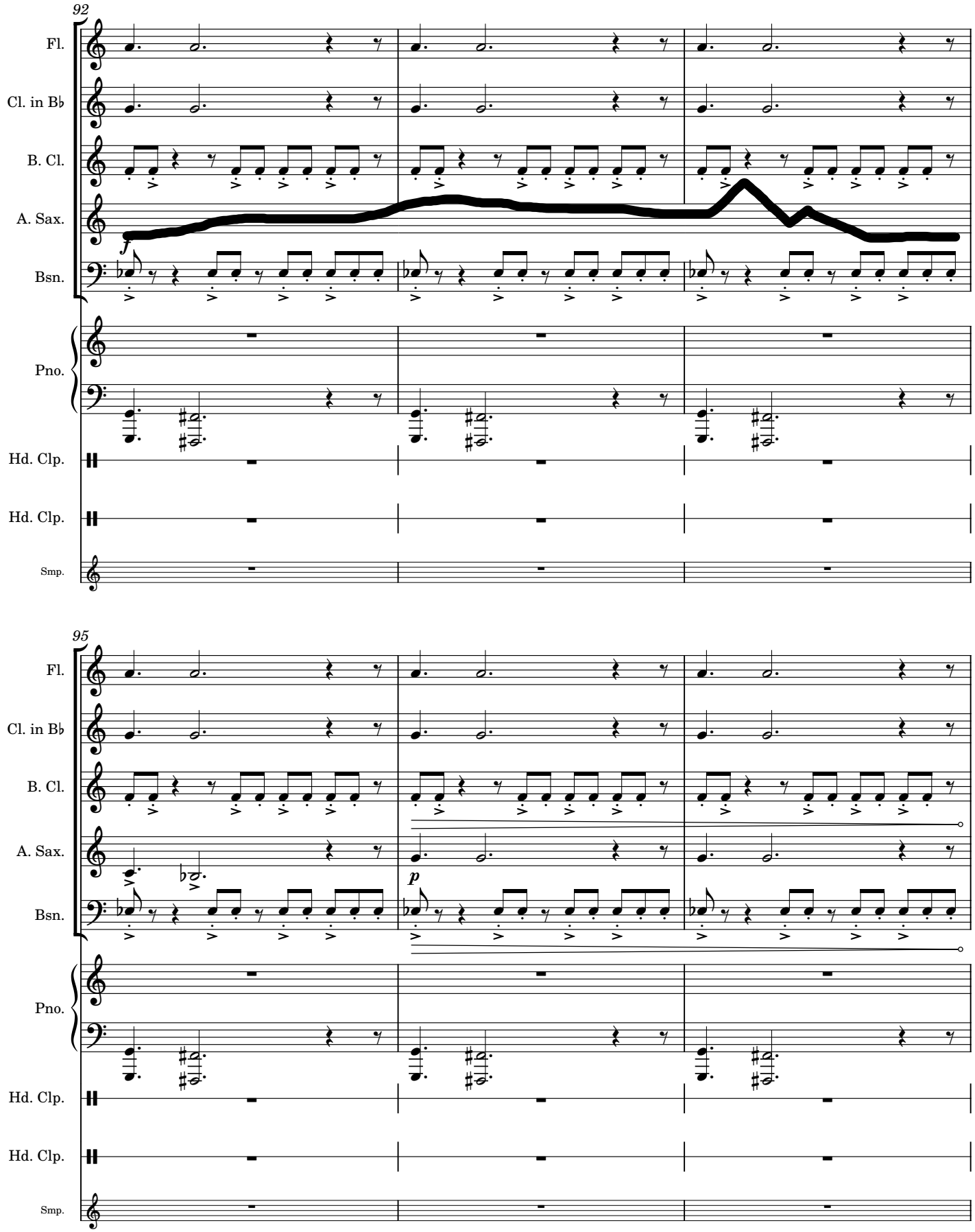

66
Distressed (♩ = 180)
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
pp
Hd. Clp.
Hd. Clp.
Smp.
68
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
mp
Hd. Clp.
Hd. Clp.
Smp.
("About 30 species per day")
("Extinction")

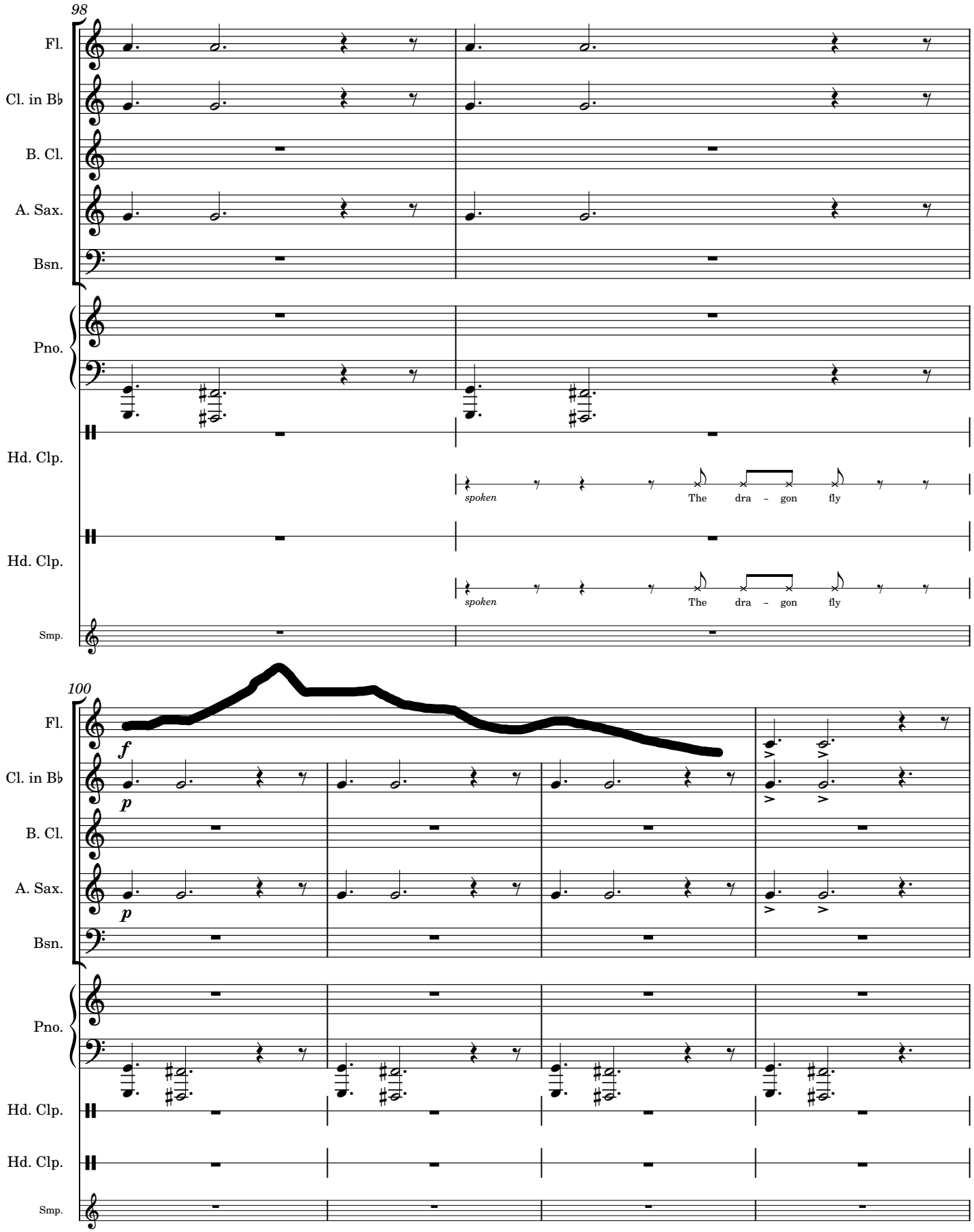

72
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
mf
mf
Pno.
Hd. Clp.
Hd. Clp.
Smp.
76
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
p
mf
p
mf
Pno.
Hd. Clp.
Hd. Clp.
Smp.
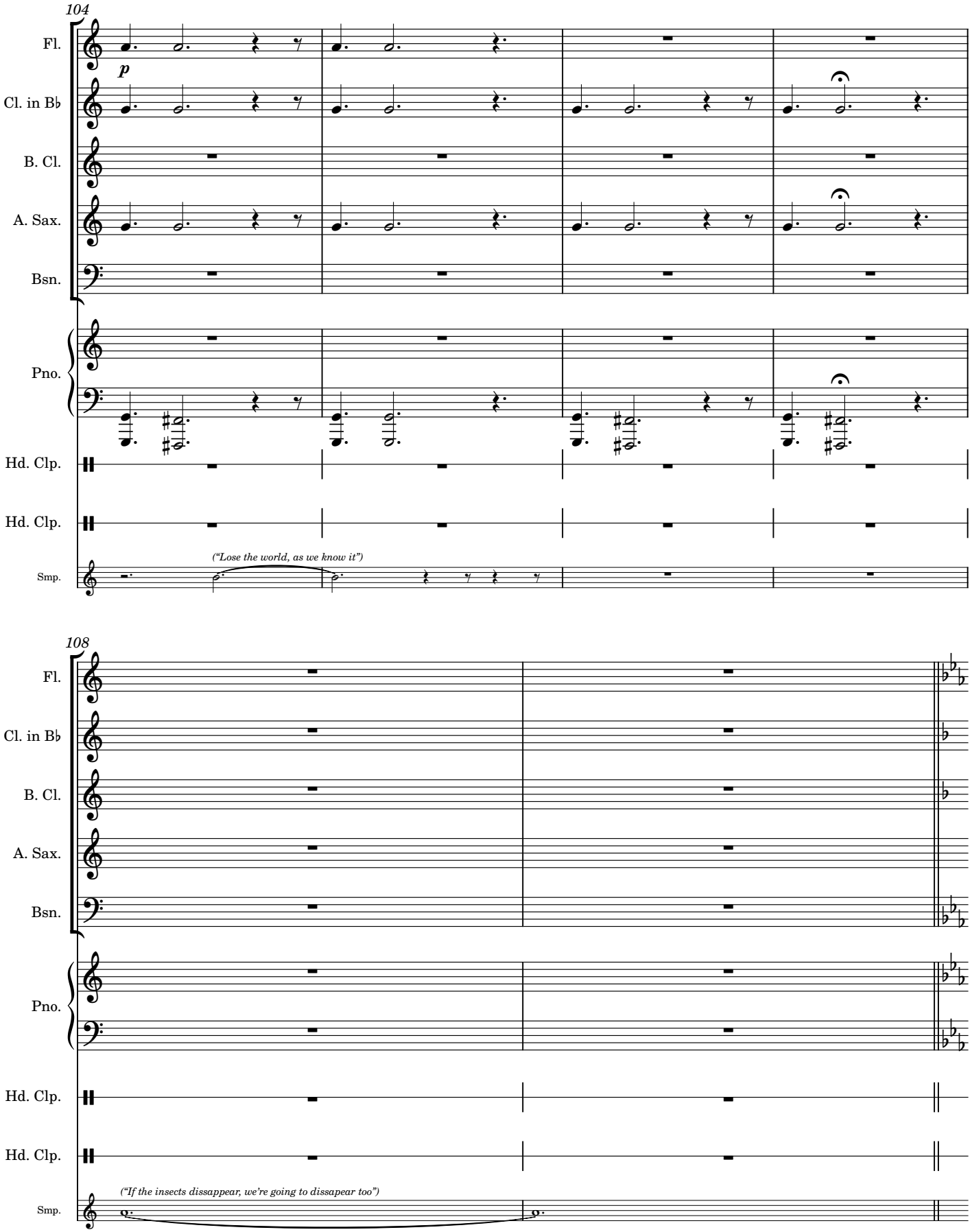

78
Fl.
mp
f
Cl. in B♭
mp
f
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
80
Fl.
p
Cl. in B♭
p
B. Cl.
A. Sax.
p
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
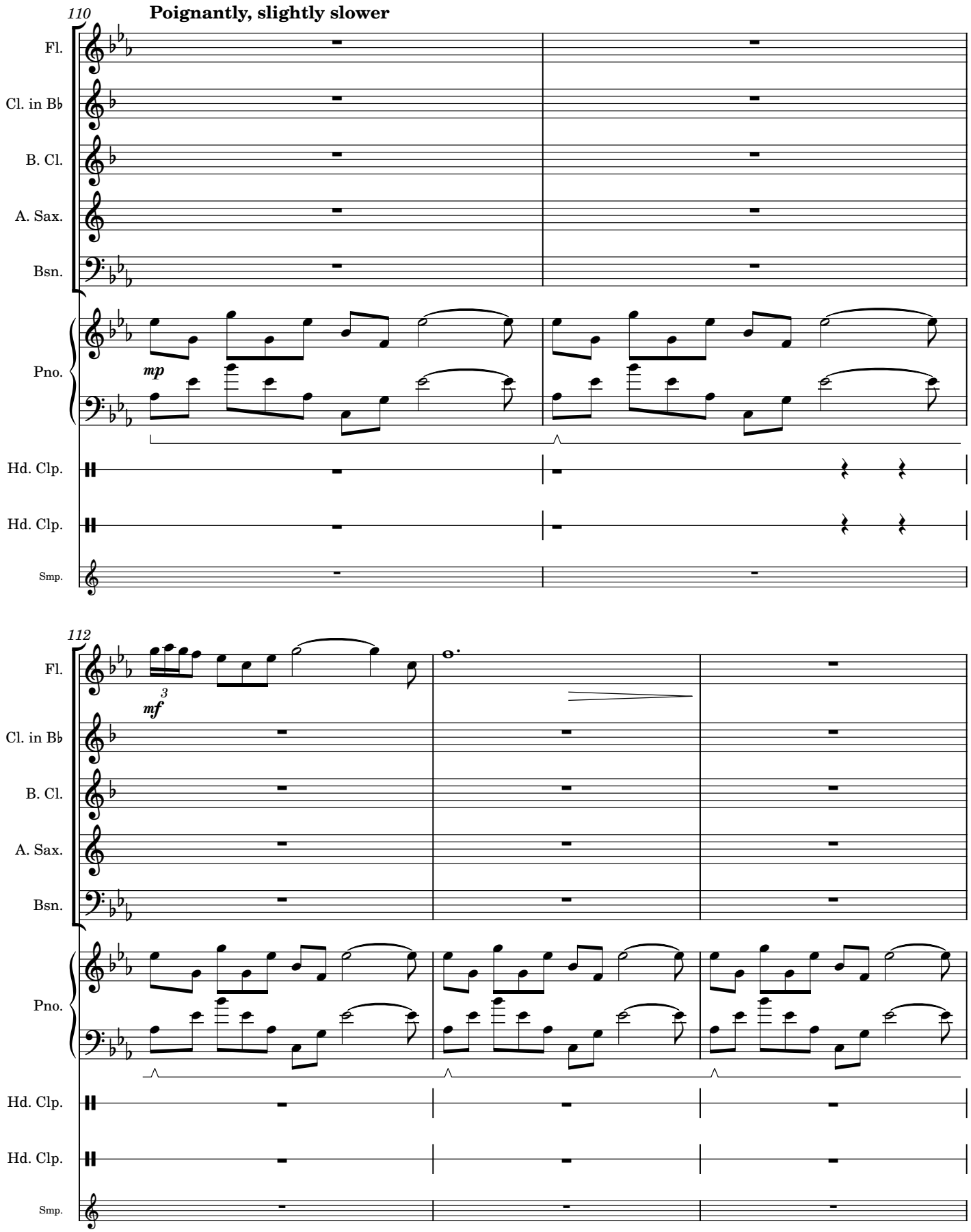

82
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
f
spoken
The Wes – tern Mon – arch
84
ppp
f
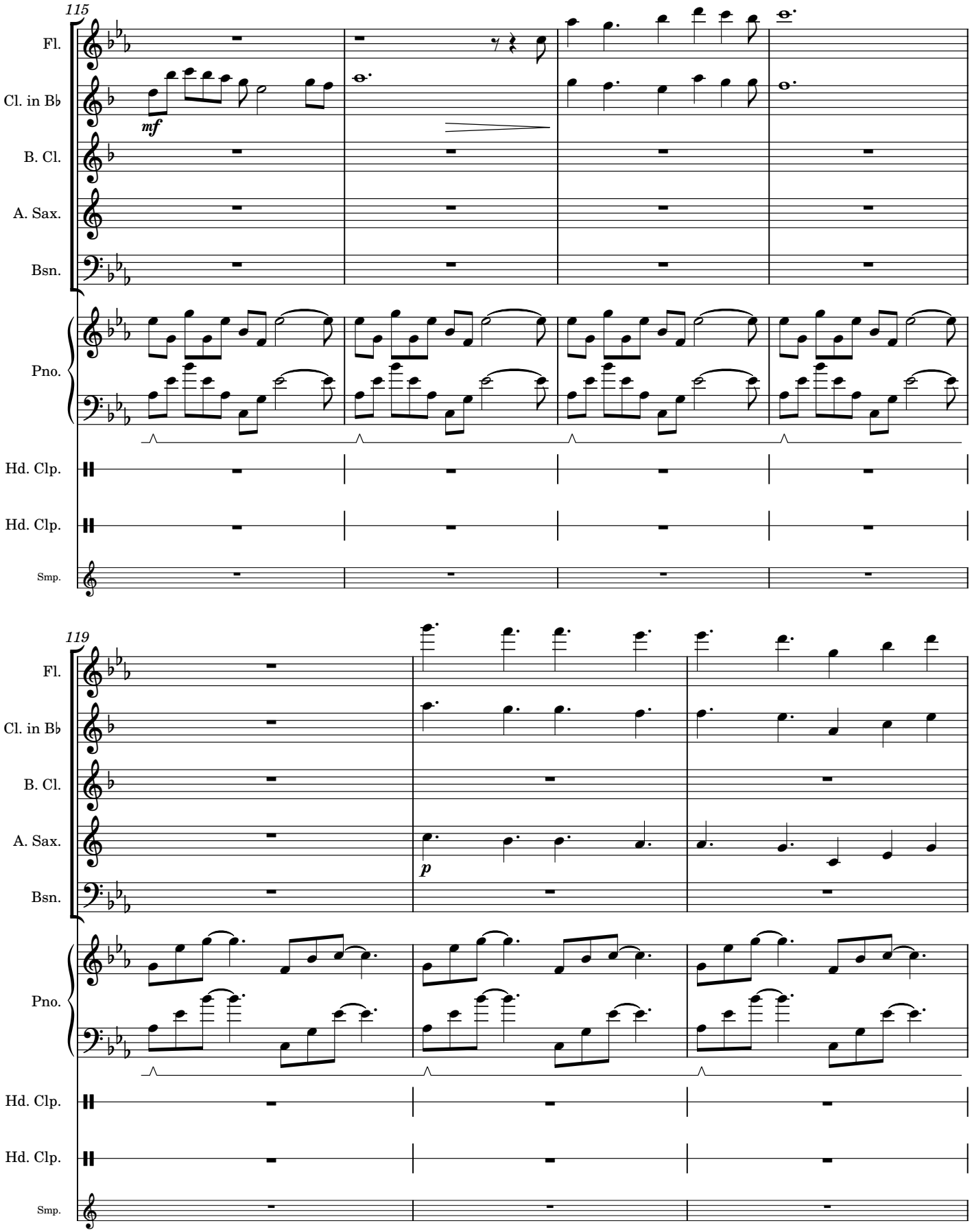

87
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
f
subito p
subito pp
90
spoken
The bum – ble bee
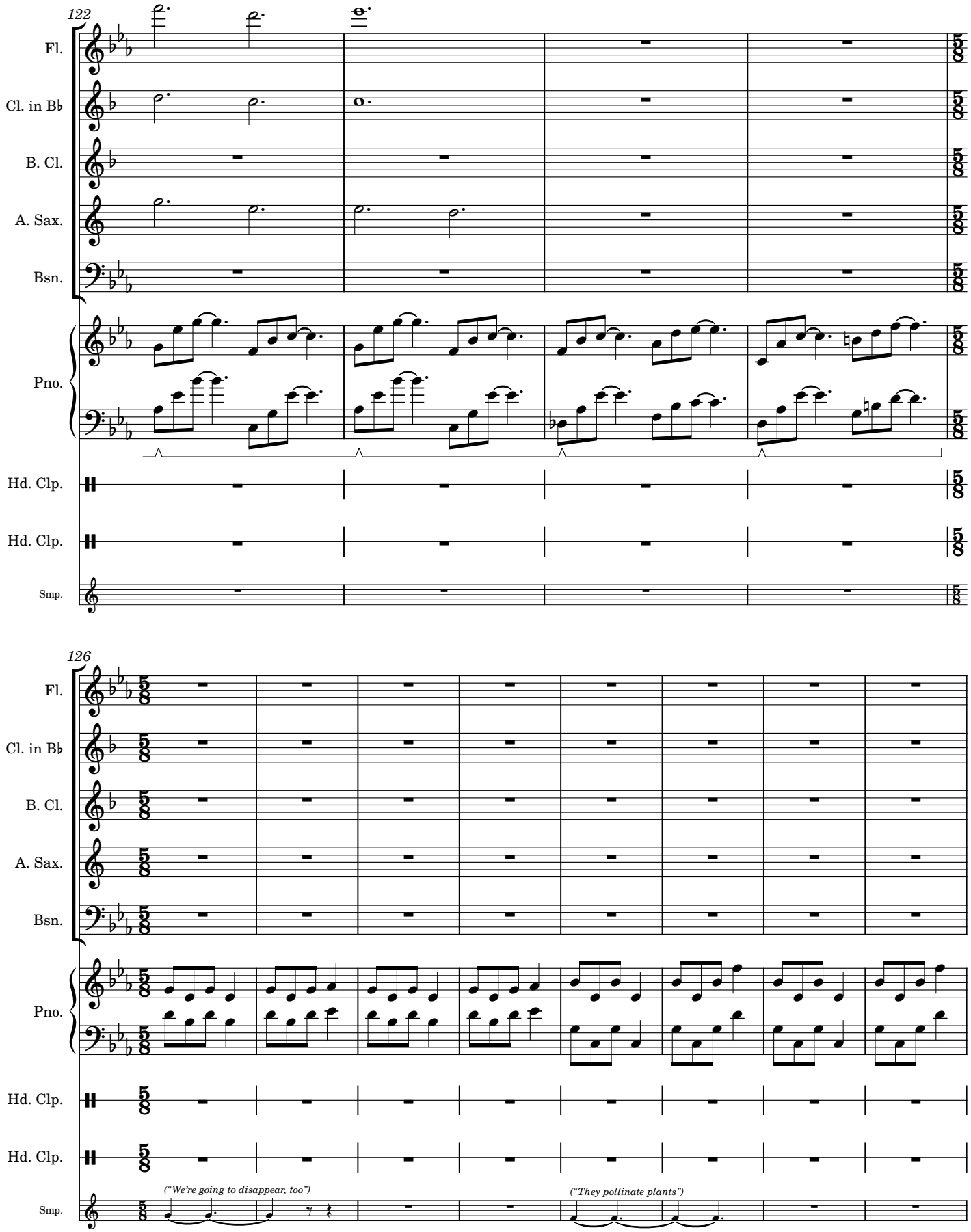

92
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
f
Pno.
Hd. Clp.
Hd. Clp.
Smp.
95
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
p
Pno.
Hd. Clp.
Hd. Clp.
Smp.
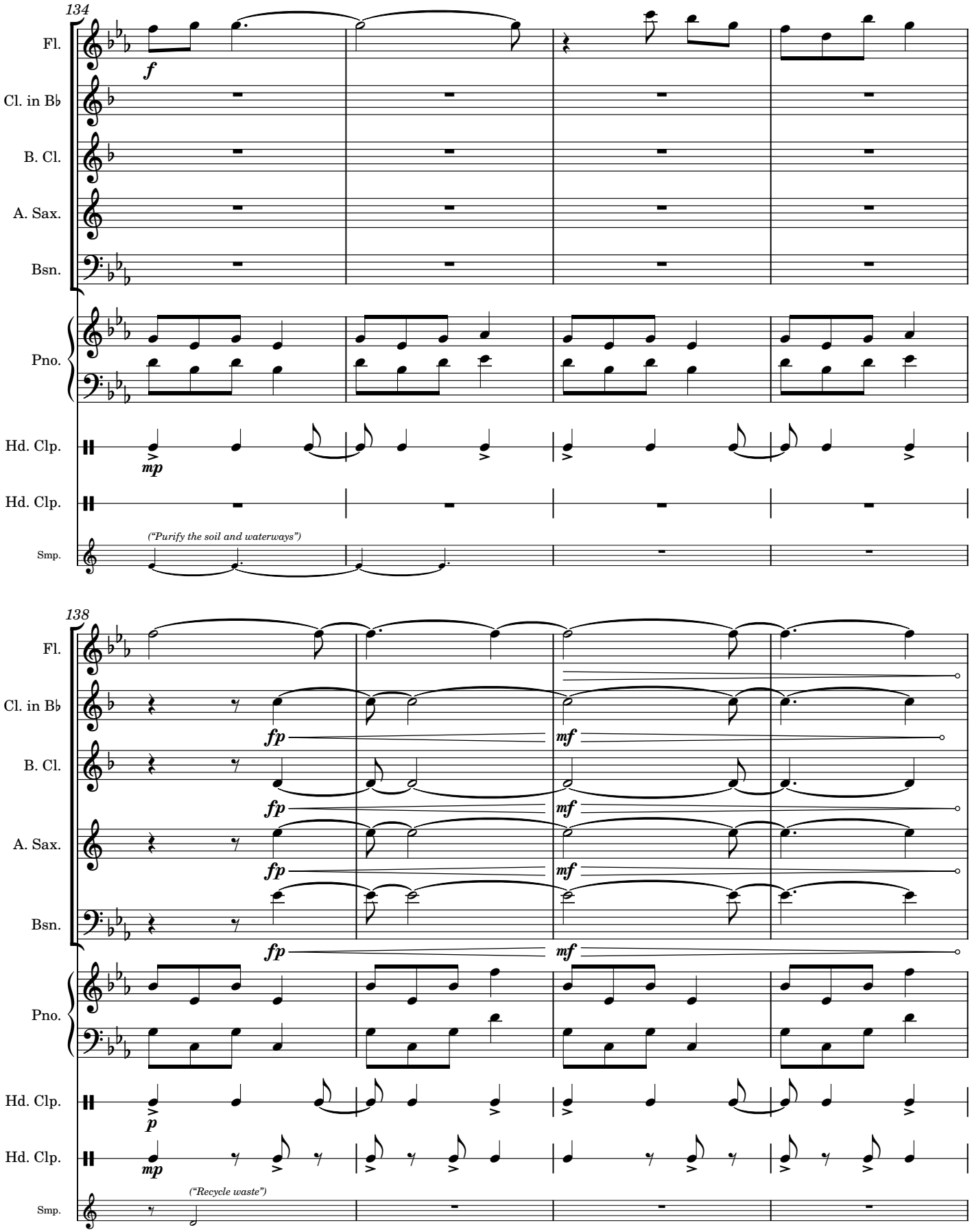

98
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
spoken
The dra – gon fly
Hd. Clp.
spoken
The dra – gon fly
Smp.
100
Fl.
f
Cl. in B♭
p
B. Cl.
A. Sax.
p
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
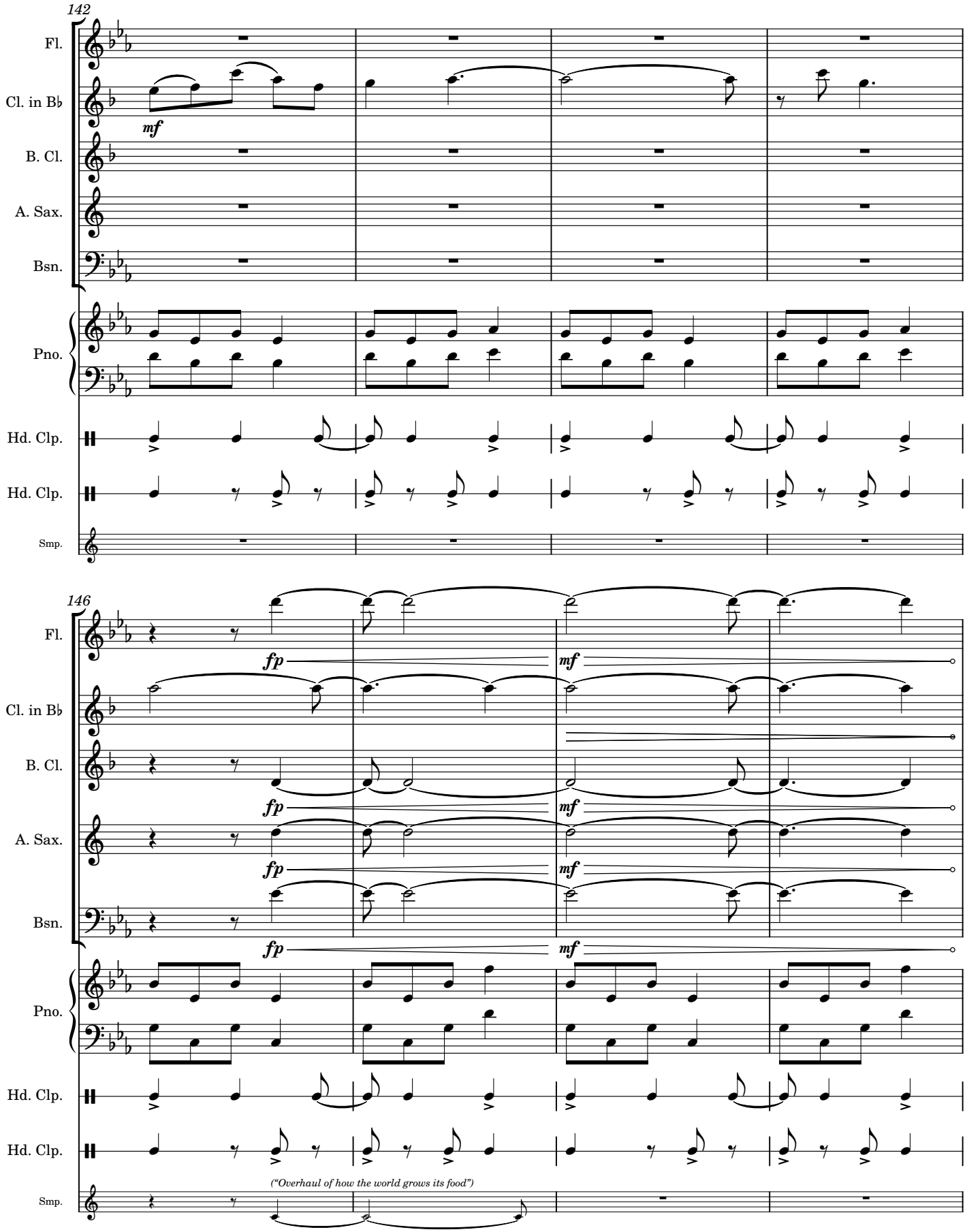

104
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
p
("Lose the world, as we know it")
108
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
("If the insects dissappear, we're going to dissapear too")
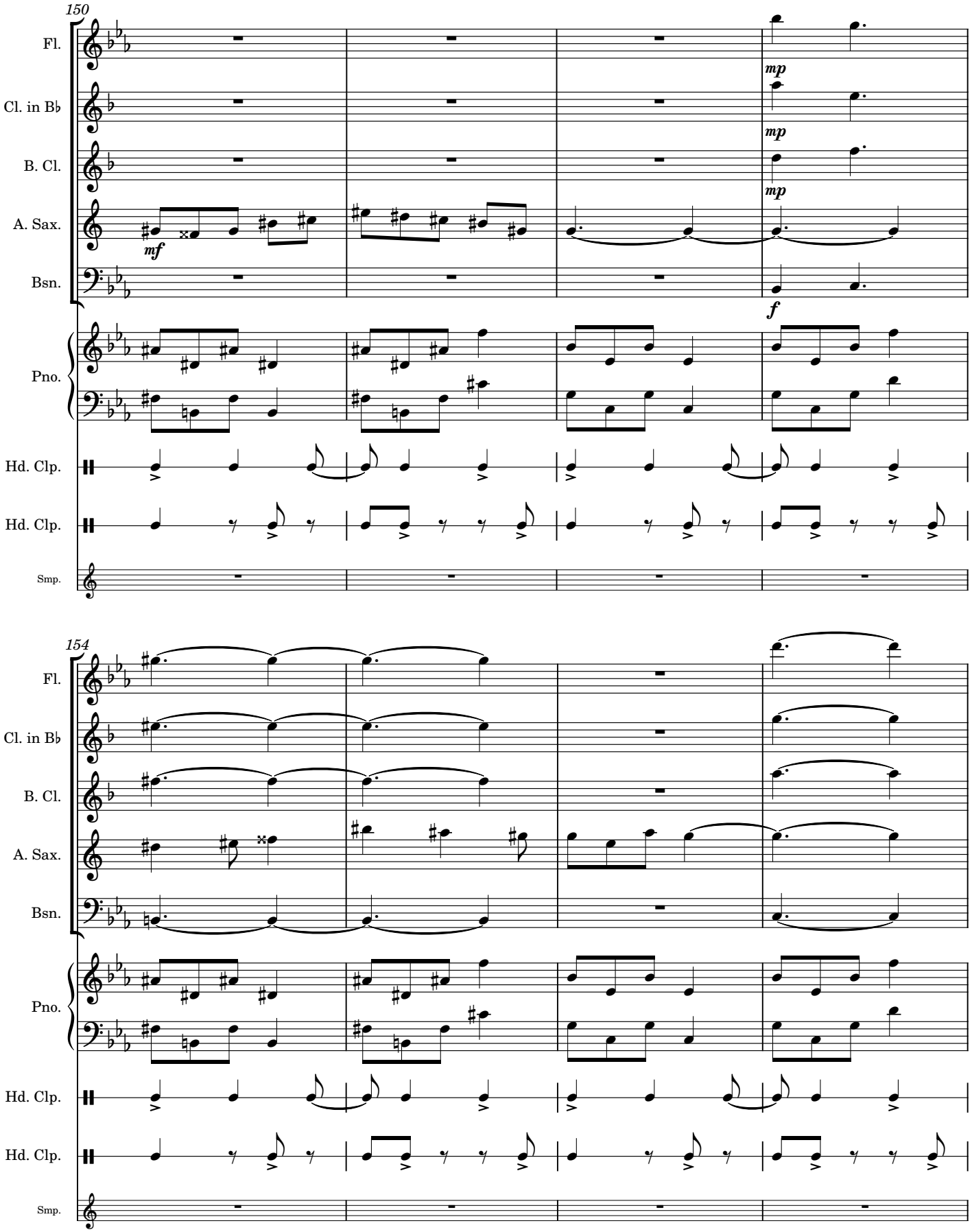

110
Poignantly, slightly slower
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
mp
Hd. Clp.
Hd. Clp.
Smp.
112
Fl.
3
mf
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
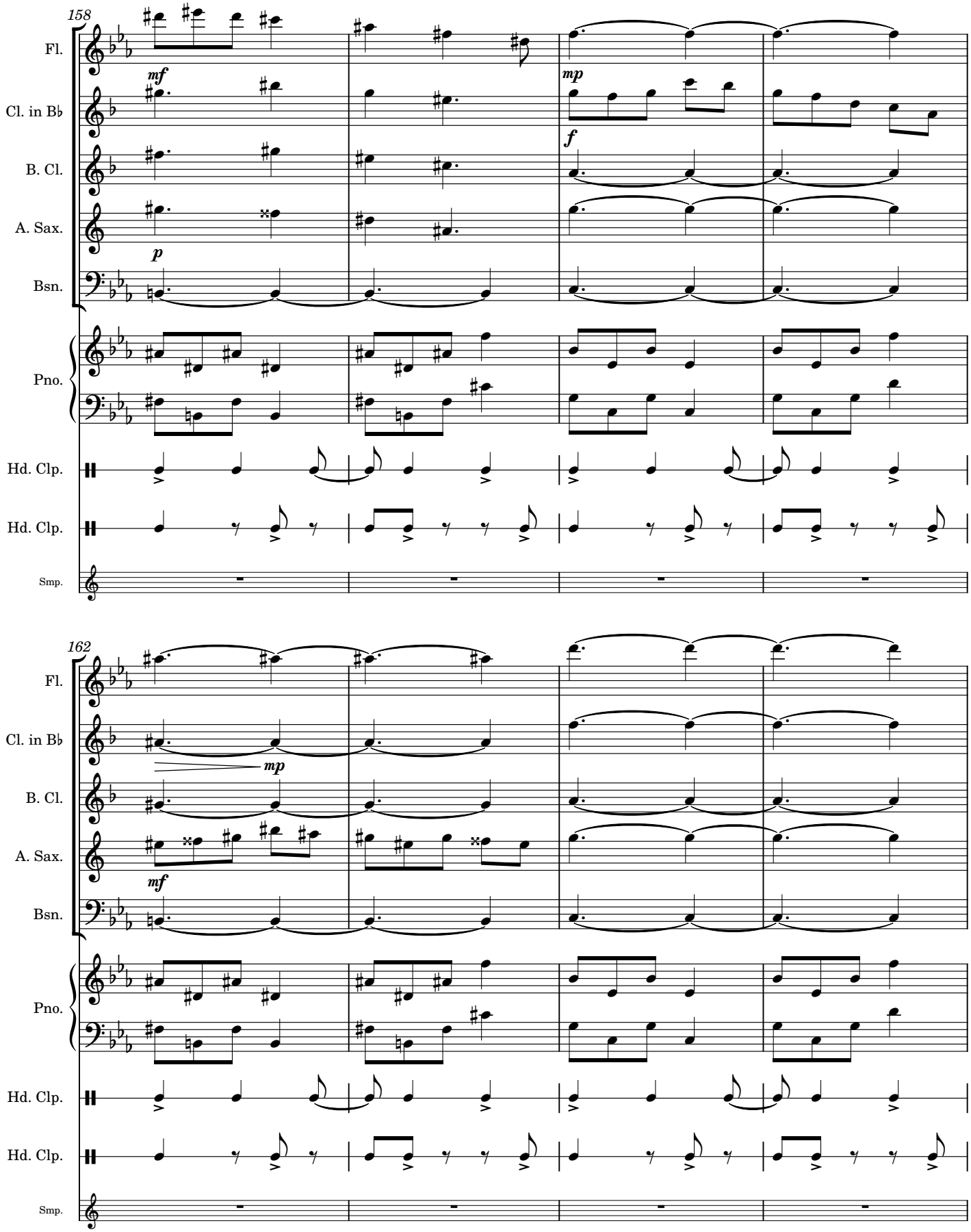

115
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
mf
Pno.
Hd. Clp.
Hd. Clp.
Smp.
119
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
p
Pno.
Hd. Clp.
Hd. Clp.
Smp.
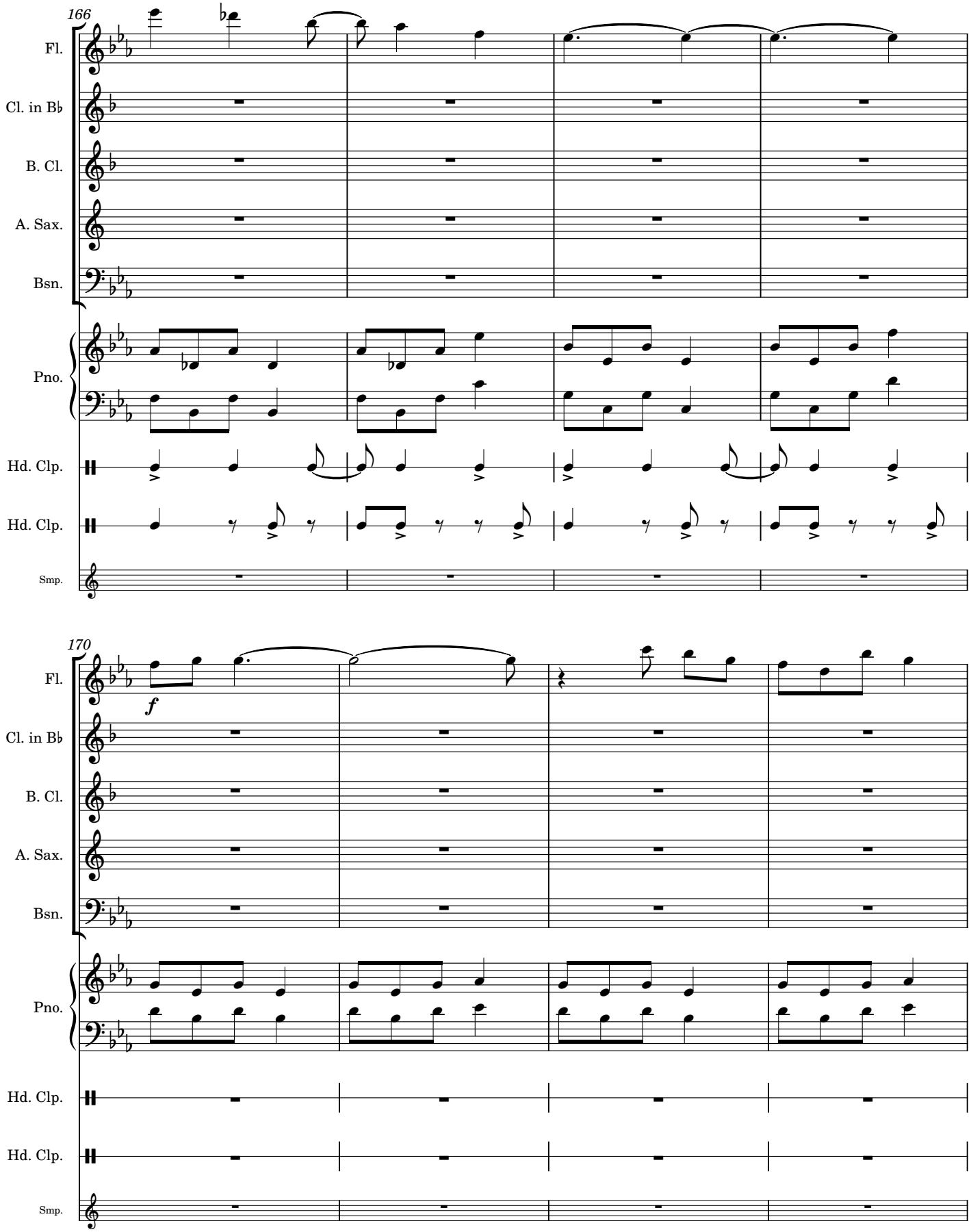

122
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
126
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
("We're going to disappear, too")
("They pollinate plants")
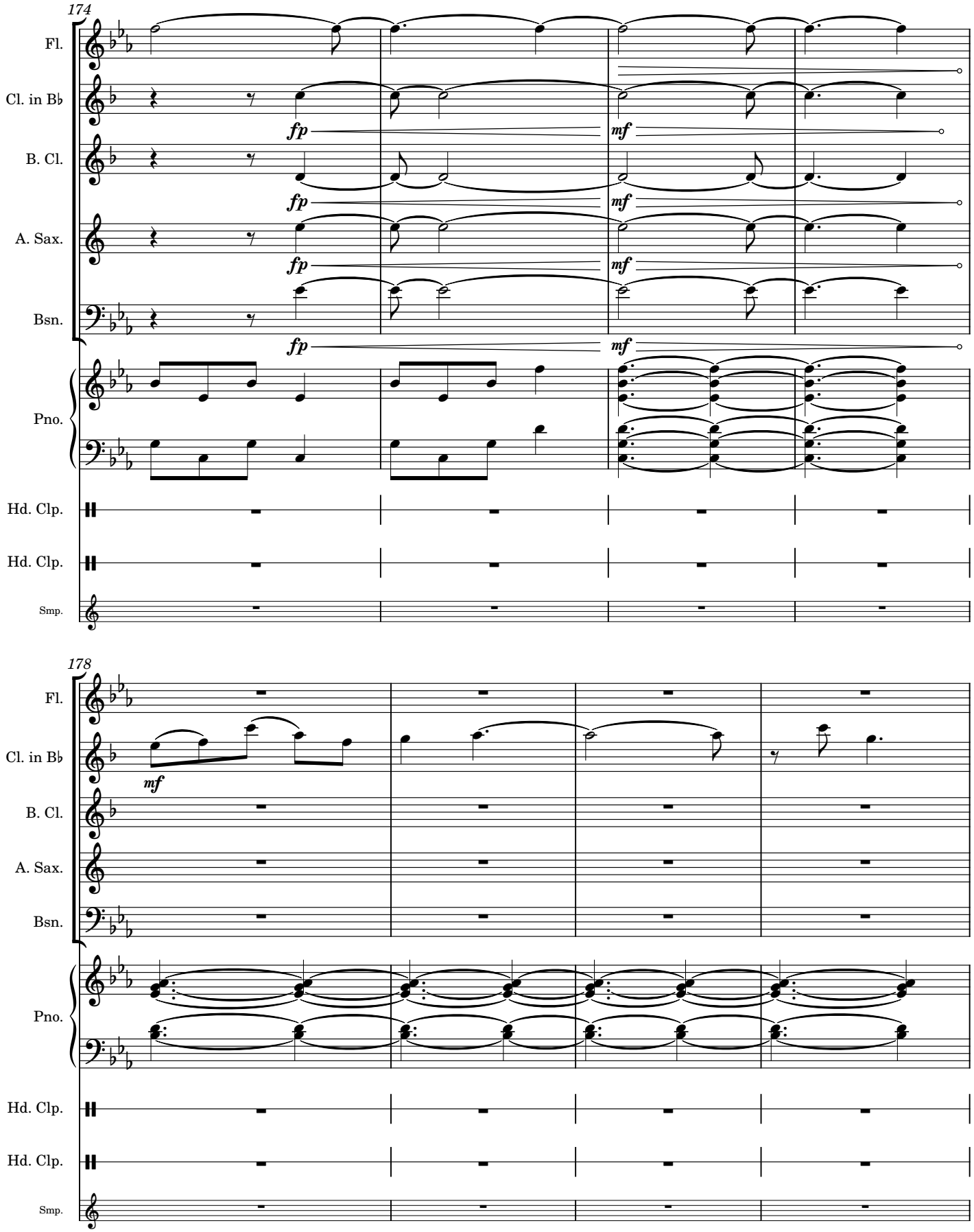

134
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
f
mp
("Purify the soil and waterways")
138
fp
mf
p
("Recycle waste")

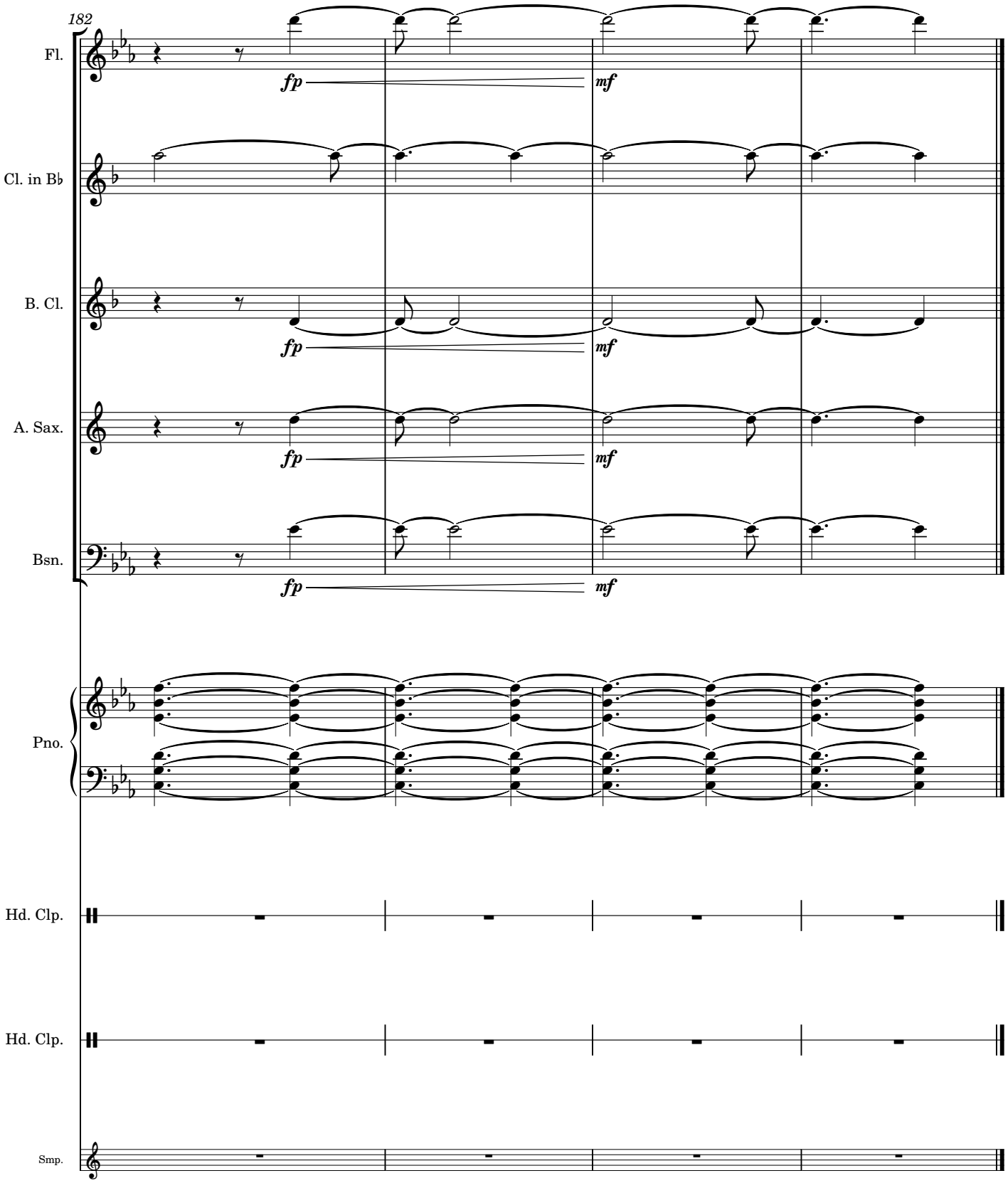

142
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mf
146
fp
mf
("Overhaul of how the world grows its food")


150
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mf
mp
mp
mp
f
154
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.


158
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mf
p
mp
f
162
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mp
mf


166
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
170
Fl.
f
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.


174
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
fp
mf
178
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
mf


182
Fl.
Cl. in B♭
B. Cl.
A. Sax.
Bsn.
Pno.
Hd. Clp.
Hd. Clp.
Smp.
fp
mf